Shield-Fi

Real-Time Financial Fraud Detection

XGBoost · SMOTE · FastAPI · Scikit-Learn
PR-AUC: 0.92 | Latency: <50ms | CPU Only

"AI-powered fraud detection built for modern payments infrastructure."

The Problem

Fraud is Expensive & Slow

Financial institutions lose over $10 Billion annually to increasingly sophisticated fraud schemes. Legacy systems can no longer keep pace with digital-first attacks.

Critical Industry Pain Points

Rising Losses

Global fraud losses are rising yearly. Consumers reported losses exceeding $10.0B in 2023 alone.

False Positives

Legitimate customers are frequently blocked by rigid rules, leading to poor UX and abandoned checkouts.

Legacy Inefficiency

Rules engines are easy to bypass and require slow manual review queues that hinder scalability.

Meet Shield-Fi: Our Solution

Real-Time ML Fraud Scoring

A plug-and-play API designed to score every transaction in milliseconds.

- **Latency:** Decision in <50ms
- **Engine:** XGBoost Risk Model
- **Action:** Approve / Review / Decline

Product Snapshot: API Endpoints

Method	Endpoint	Use Case
GET	/health	System uptime and health check
GET	/model/info	Current model version and performance metrics
POST	/score	Real-time single transaction fraud scoring
POST	/score/batch	High-throughput batch scoring (up to 100 txns)

Ideal for: Checkout flows, Digital Wallets, Neobanks, and BNPL platforms.

Unrivaled Detection Accuracy

0.92

PR-AUC Score

Precision Where It Matters

In datasets where fraud is rare (0.17% rate), traditional accuracy is a vanity metric. Shield-Fi focuses on **PR-AUC** to ensure we catch fraud without annoying legitimate users.

$$\text{PR-AUC} = \int_0^1 p(r)\,dr$$

Abstract Technology Network Server

Secret Sauce: Engineered Signals

- **Velocity Spikes:** Monitoring 5m / 1hr / 1d transaction frequency.
- **Behavioral Z-Scores:** Anomaly detection against user baselines.
- **Cross-Border Logic:** Intelligent travel-pattern recognition.
- **Merchant Risk:** Real-time indexing of high-risk endpoints.

Result: +15% Recall vs Raw Feature Baselines.

The Competitive Edge

Shield-Fi out-performs legacy rule-based engines in both speed and accuracy while reducing cloud compute costs.

Market Strategy

Revenue Model

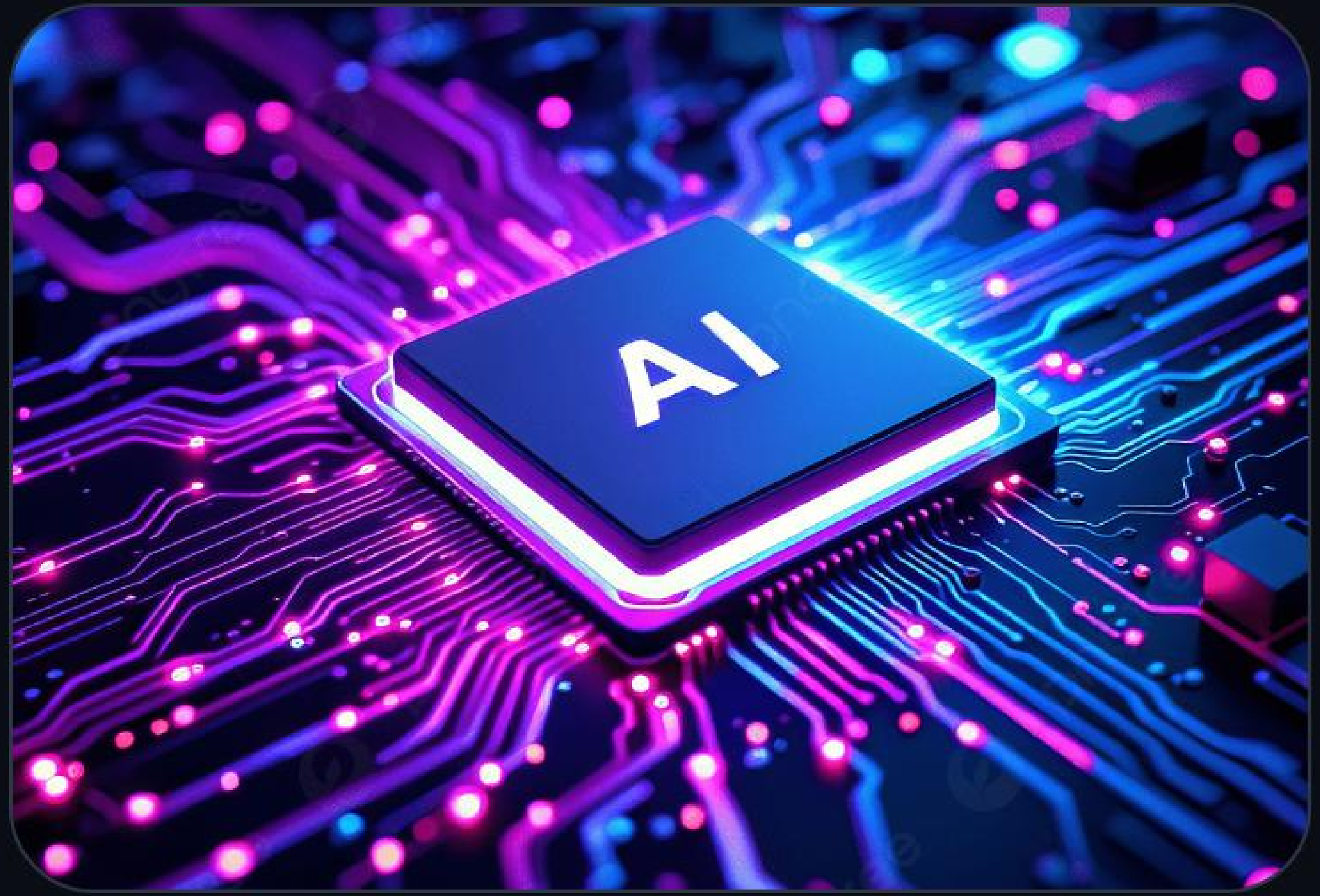
- Usage-based API Pricing
- Monthly Enterprise Tiers
- Analytics Dashboard Upsell

Target Verticals

- Payment Processors
- Neobanks & Wallets
- BNPL Platforms

Why Now? Market Timing

Digital payment volume hitting record highs annually.



Mainstream AI adoption and GPU-less ML maturity.

Increasing regulatory pressure for robust AML/Fraud compliance.

Vision: Future Roadmap

Q3 2024
Device Fingerprinting & Latency Optimization

Q1 2025
Graph Analytics for Fraud Network Detection

Q3 2025
Explainable AI (XAI) for Regulatory Audits

2026+
Self-Learning Automated Retraining Loop

Shield-Fi

Modern fraud prevention for real-time finance.

Our Ask

Pilot partners · Fintech integrations · Strategic funding

www.shield-fi.io | contact@shield-fi.io

Image Sources

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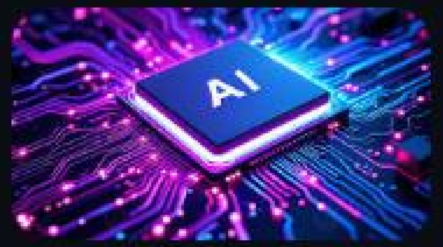

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